Ahmadreza Nourozi
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Schlossplatz 4, 91054 Erlangen, Germany
Ahmadreza.nourozi@fau.de

October 20, 2026

The Editors
Computers in Biology and Medicine

Dear Editors,

We submit the Full Length Article "Separating decision-rule failures from representation failures in external validation of diabetic retinopathy grading models" for consideration in *Computers in Biology and Medicine*, section Foundations of Artificial Intelligence in Medicine.

When a diabetic retinopathy grading model fails on a new population, the failure is usually attributed to the learned representation and answered by changing the model. Ordinal grading models, however, convert scores into grades with a decision rule that is rarely examined. Across ten development models, four external populations (APTOS 2019, DDR, DeepDRiD and SUSTech-SYSU), the reverse direction of transfer, two models trained on about 28 000 images and five training seeds, we show that the ranking of images transferred while the mapping from score to grade did not: the same models over-graded one population and under-graded another, so that a referral threshold fitted for 85% sensitivity reached 99.33% on one and 63.38% on the other. Refitting cut-points alone recovered part of the loss, but only where ranking survived and only with local labels; label-free adaptation did not help. We report with equal weight that two of our own predictions were falsified.

We believe the work suits the journal for three reasons. It contributes evaluation methodology for medical AI: a decomposition of external performance, and a matched decision rule for comparing models, with its record of changed and unchanged conclusions, including a published model whose reported advantage did not survive it. It is directly usable in deployment: we quantify how many locally labelled images a referral threshold needs and how often that procedure fails, and translate the operating point into missed referrals per 10 000 people screened. And it is reproducible: the analysis plans were committed to a public repository before the corresponding analyses, every number in the manuscript is generated from archived per-image predictions, and code, splits and predictions are public.

The manuscript has not been published and is not under consideration elsewhere. Parts of the development work form the basis of a Master's thesis at the University of Tehran. All data are public and de-identified. The authors declare no competing interests. AI-assisted coding and editing are disclosed in the Methods and in the generative-AI declaration.

Sincerely,

Ahmadreza Nourozi
(on behalf of both authors)